

**LOMBA KOMPETENSI SISWA
PENDIDIKAN MENENGAH
TINGKAT PROVINSI
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**TEST PROJECT
MODUL B – NETWORK SYSTEMS
IT NETWORK SYSTEMS ADMINISTRATION**

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Introduction

In today's IT landscape, proficiency in network technology is increasingly vital for individuals aspiring to excel in any IT engineering discipline. This test project presents numerous challenges drawn from real-world scenarios, focusing predominantly on IT Networking and Integration. Successfully completing this project with a high score demonstrates your readiness to manage network infrastructures for multi-branch enterprises.

Description of project and tasks

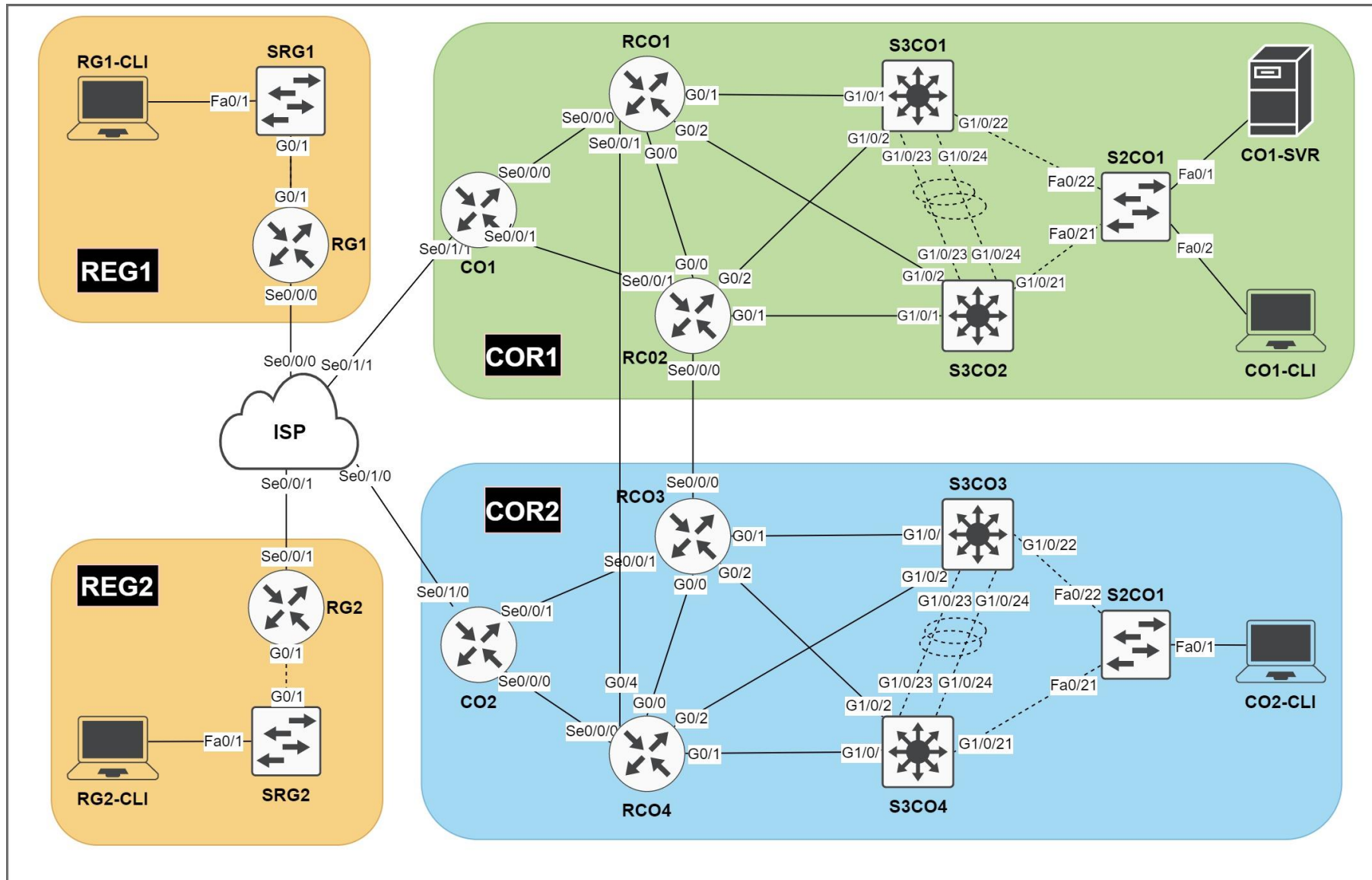
This test project is designed using a variety of network technologies that should be familiar to those who have studied for Cisco CCNA - Implementing and Administering Cisco Solutions certification track.

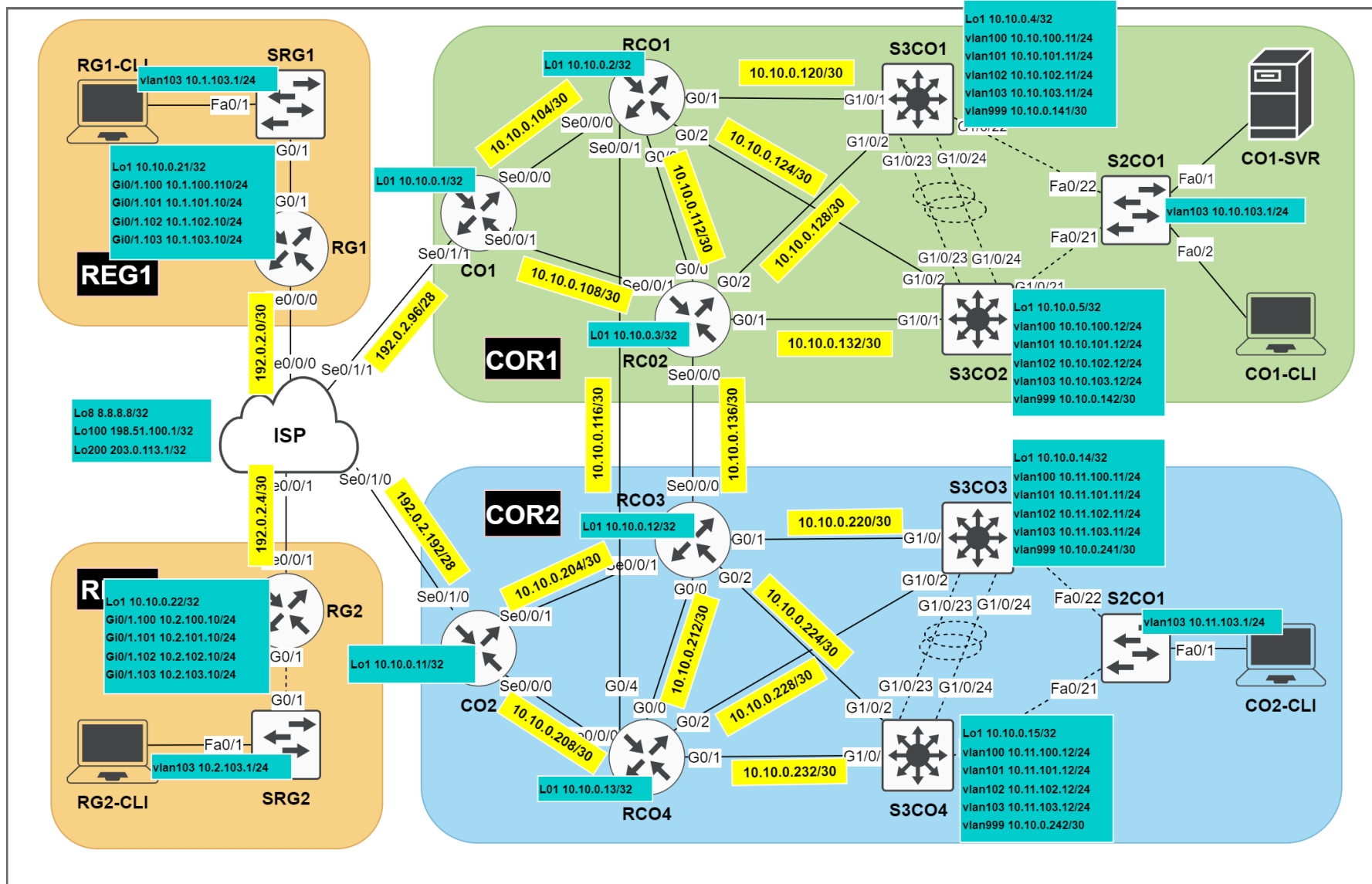
Configuration tasks are broken down into seven areas given below expanded in section Instructions to the Competitors.

- Basic configuration
- L2 services
- OSPF
- EIGRP
- IP Services
- Security

Physical Topology

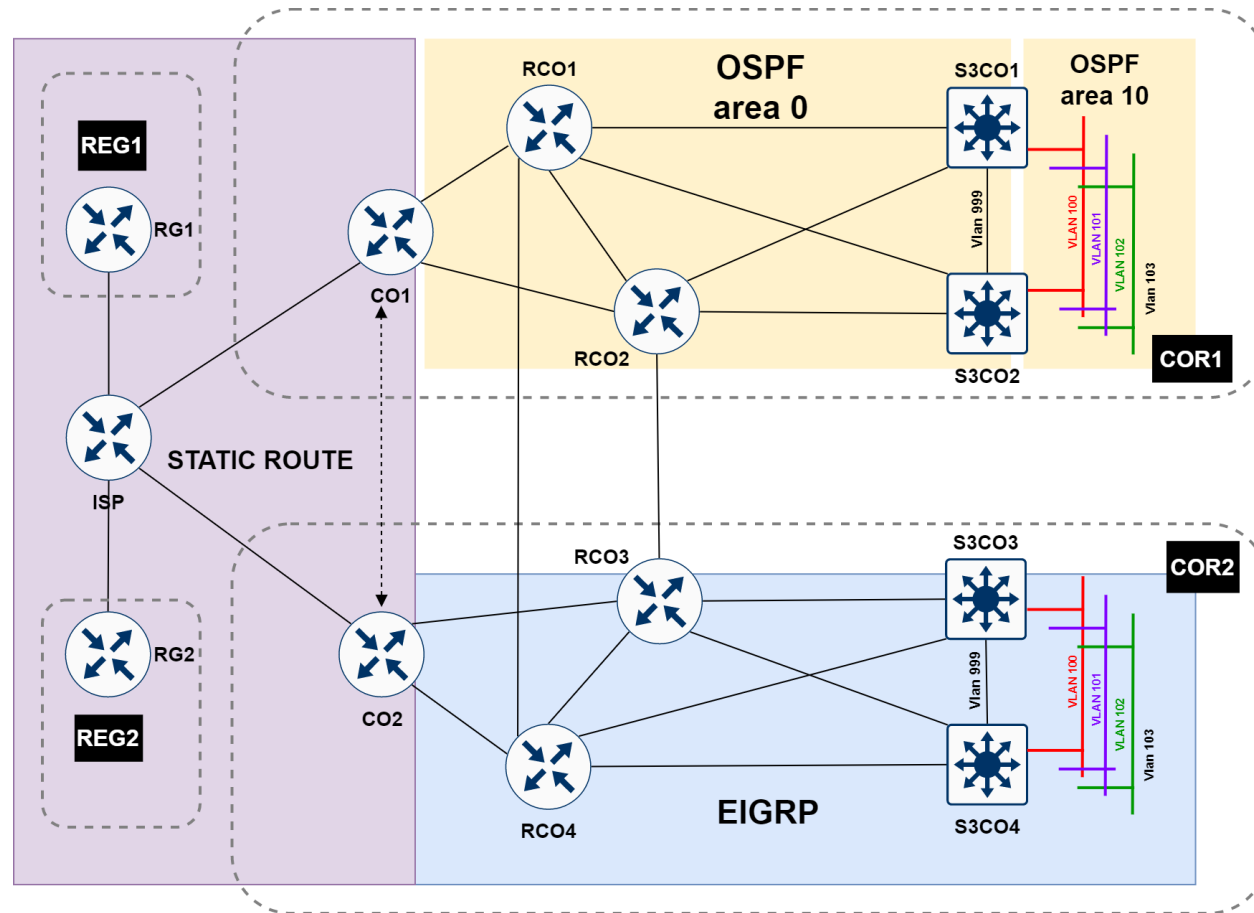
Below is the physical topology of the Test Project.





Routing Topology

BGP, EIGRP, and OSPF Routing topology is shown below.



Site	VLAN100	VLAN101	VLAN102	VLAN103
COR1	10.10.100.0	10.10.101.0	10.10.102.0	10.10.103.0
COR2	10.11.100.0	10.11.101.0	10.11.102.0	10.11.103.0
REG1	10.1.100.0	10.1.101.0	10.1.102.0	10.1.103.0
REG2	10.2.100.0	10.2.101.0	10.2.102.0	10.2.103.0

Instructions to the Competitor

Please carefully read below instruction.

1. Please do not modify ISP1's configuration. All ISP1 configurations are completed, and you are not supposed to touch that device.
2. Read all tasks in each section before proceeding with any configuration. The completion of any item may require the completion of any previous or later item.
3. If you need to add a command to ensure the service functions, there will be no point deduction. You are free to add commands, the most important thing is that the main objective of the task can function properly.
4. Save your configurations frequently; accidents do and will happen.
5. All Router and Switch IP Addresses configuration already done for you **EXCEPT IP ADDRESS OF INTERFACE VLAN**.

DEVICE	INTERFACE	IPV4 ADDRESS	NEIGHBOR
ISP1	Serial0/1/1	192.0.2.101/28	CO1
ISP1	Serial0/1/0	192.0.2.201/28	CO2
ISP1	Serial0/0/0	192.0.2.1/30	RG1
ISP1	Serial0/0/1	192.0.2.5/30	RG2
ISP1	Loopback100	198.51.100.1/24	N/A
ISP1	Loopback200	203.0.113.1/24	N/A
CO1	Serial0/0/0	10.10.0.105/30	RCO1
CO1	Serial0/0/1	10.10.0.109/30	RCO2
CO1	Serial0/1/1	192.0.2.102/28	ISP1
CO1	Loopback1	10.10.0.1/32	N/A
CO2	Serial0/0/1	10.10.0.205/30	RCO3
CO2	Serial0/0/0	10.10.0.209/30	RCO4
CO2	Serial0/1/0	192.0.2.202/28	ISP1
CO2	Loopback1	10.10.0.11/32	N/A
RCO1	GigabitEthernet0/0	10.10.0.113/30	RCO2
RCO1	GigabitEthernet0/1	10.10.0.121/30	S3CO1
RCO1	GigabitEthernet0/2	10.10.0.125/30	S3CO2
RCO1	Serial0/0/0	10.10.0.106/30	CO1
RCO1	Serial0/0/1	10.10.0.117/30	RCO4
RCO1	Loopback1	10.10.0.2/32	N/A
RCO2	GigabitEthernet0/0	10.10.0.114/30	RCO1
RCO2	GigabitEthernet0/1	10.10.0.133/30	S3CO2
RCO2	GigabitEthernet0/2	10.10.0.129/30	S3CO1
RCO2	Serial0/0/1	10.10.0.110/30	CO1
RCO2	Serial0/0/0	10.10.0.137/30	RCO3
RCO2	Loopback1	10.10.0.3/32	N/A

DEVICE	INTERFACE	IPV4 ADDRESS	NEIGHBOR
RCO3	GigabitEthernet0/0	10.10.0.213/30	RCO4
RCO3	GigabitEthernet0/1	10.10.0.221/30	S3CO3
RCO3	GigabitEthernet0/2	10.10.0.225/30	S3CO4
RCO3	Serial0/0/1	10.10.0.206/30	CO2
RCO3	Serial0/0/0	10.10.0.138/30	RCO2
RCO3	Loopback1	10.10.0.12/32	N/A
RCO4	GigabitEthernet0/0	10.10.0.214/30	RCO3
RCO4	GigabitEthernet0/1	10.10.0.233/30	S3CO4
RCO4	GigabitEthernet0/2	10.10.0.229/30	S3CO3
RCO4	Serial0/0/0	10.10.0.210/30	CO2
RCO4	Serial0/0/1	10.10.0.118/30	RCO1
RCO4	Loopback1	10.10.0.13/32	N/A
S3CO1	GigabitEthernet0/1	10.10.0.122/30	RCO1
S3CO1	GigabitEthernet0/2	10.10.0.130/30	RCO2
S3CO1	Loopback1	10.10.0.4/32	N/A
S3CO2	GigabitEthernet0/1	10.10.0.134/30	RCO2
S3CO2	GigabitEthernet0/2	10.10.0.126/30	RCO1
S3CO2	Loopback1	10.10.0.5/32	N/A
S3CO3	GigabitEthernet0/1	10.10.0.222/30	RCO3
S3CO3	GigabitEthernet0/2	10.10.0.230/30	RCO4
S3CO3	Loopback1	10.10.0.14/32	N/A
S3CO4	GigabitEthernet0/1	10.10.0.234/30	RCO4
S3CO4	GigabitEthernet0/2	10.10.0.226/30	RCO3
S3CO4	Loopback1	10.10.0.15/32	N/A
S2CO1	Vlan103	10.10.103.1/24	N/A
S2CO2	Vlan103	10.11.103.1/24	N/A
RG1	Serial0/0/0	192.0.2.2/30	ISP1
RG1	GigabitEthernet0/1.100	10.1.100.10/24	N/A
RG1	GigabitEthernet0/1.101	10.1.101.10/24	N/A
RG1	GigabitEthernet0/1.102	10.1.102.10/24	N/A
RG1	GigabitEthernet0/1.103	10.1.103.10/24	N/A
RG1	Loopback1	10.10.0.21/32	N/A
SRG1	Vlan103	10.1.103.1/24	N/A
RG2	Serial0/0/1	192.0.2.6/30	ISP1
RG2	GigabitEthernet0/1.100	10.2.100.10/24	N/A

DEVICE	INTERFACE	IPV4 ADDRESS	NEIGHBOR
RG2	GigabitEthernet0/1.101	10.2.101.10/24	N/A
RG2	GigabitEthernet0/1.102	10.2.102.10/24	N/A
RG2	GigabitEthernet0/1.103	10.2.103.10/24	N/A
RG2	Loopback1	10.10.0.22/32	N/A
SRG2	Vlan103	10.2.103.1/24	N/A

6. IPv6 addresses in COR2 devices.

DEVICE	INTERFACE	IPV4 ADDRESS	NEIGHBOR
CO2	Serial0/0/1	2001:DB8:0:204::1/64	RCO3
CO2	Serial0/0/0	2001:DB8:0:208::1/64	RCO4
CO2	Serial0/1/0	2001:DB8:0:200::2/64	ISP1
CO2	Loopback1	2001:DB8:0:11::1/64	N/A
RCO3	GigabitEthernet0/0	2001:DB8:0:212::1/64	RCO4
RCO3	GigabitEthernet0/1	2001:DB8:0:220::1/64	S3CO3
RCO3	GigabitEthernet0/2	2001:DB8:0:224::1/64	S3CO4
RCO3	Serial0/0/1	2001:DB8:0:204::2/64	CO2
RCO3	Loopback1	2001:DB8:0:12::1/64	N/A
RCO4	GigabitEthernet0/0	2001:DB8:0:212::2/64	RCO3
RCO4	GigabitEthernet0/1	2001:DB8:0:232::1/64	S3CO4
RCO4	GigabitEthernet0/2	2001:DB8:0:228::1/64	S3CO3
RCO4	Serial0/0/0	2001:DB8:0:208::2/64	CO2
RCO4	Loopback1	2001:DB8:0:13::1/64	N/A
S3CO3	GigabitEthernet0/1	2001:DB8:0:220::2/64	RCO3
S3CO3	GigabitEthernet0/2	2001:DB8:0:228::2/64	RCO4
S3CO3	Loopback1	2001:DB8:0:14::1/64	N/A
S3CO4	GigabitEthernet0/1	2001:DB8:0:232::2/64	RCO4
S3CO4	GigabitEthernet0/2	2001:DB8:0:224::2/64	RCO3
S3CO4	Loopback1	2001:DB8:0:15::1/64	N/A

Configuration Tasks

Basic Configurations

1. Configure hostnames for all devices in the topology.
2. Configure all routers and switches in COR1, COR2, REG1 and REG2 to use WIB timezone.
3. Configure domain name **lks2026.id** for all devices in the topology.
4. Configure secret password **Skill39@ID** in all routers and switches. Also define local username **admin** with secret **Skill39@ID** in all routers and switches. That user should have the privilege **15**.
5. Configure IPv4 and IPv6 VLAN Interface in

DEVICE	INTERFACE	IPv4 ADDRESS	NEIGHBOR
S3CO1	Vlan100	10.10.100.11/24	N/A
	Vlan101	10.10.101.11/24	N/A
	Vlan102	10.10.102.11/24	N/A
	Vlan103	10.10.103.11/24	N/A
	Vlan999	10.10.0.141/30	S3CO2
S3CO2	Vlan100	10.10.100.12/24	N/A
	Vlan101	10.10.101.12/24	N/A
	Vlan102	10.10.102.12/24	N/A
	Vlan103	10.10.103.12/24	N/A
	Vlan999	10.10.0.142/30	S3CO2
S3CO3	Vlan100	2001:DB8:11:100::11/64	N/A
		10.11.100.11/24	N/A
	Vlan101	2001:DB8:11:101::11/64	N/A
		10.11.101.11/24	N/A
	Vlan102	2001:DB8:11:102::11/64	N/A
		10.11.102.11/24	N/A
	Vlan103	10.11.103.11/24	N/A
	Vlan999	2001:DB8:0:240::1/64	S3CO4
		10.10.0.241/30	S3CO4
S3CO4	Vlan100	2001:DB8:11:100::12/64	N/A
		10.11.100.12/24	N/A
	Vlan101	2001:DB8:11:101::12/64	N/A
		10.11.101.12/24	N/A
	Vlan102	2001:DB8:11:102::12/64	N/A
		10.11.102.12/24	N/A
	Vlan103	10.11.103.12/24	N/A
	Vlan999	2001:DB8:0:240::2/64	S3CO3
		10.10.0.242/30	S3CO3

L2 Services

1. Configure VTP domain “**LKS2026**” on S3CO1, S3CO2, S3CO3, S3CO4, S2CO1, S2CO2. Configure S3CO1 and S3CO3 as VTP Server and other Switches as clients.
2. Configure following VLANs on S3CO1, S3CO3, SRG1 and SRG2

VLAN ID	VLAN NAME	DEVICES
100	SERVER	S3CO1, S3CO3, SRG1, SRG2
101	CLIENT_1	S3CO1, S3CO3, SRG1, SRG2
102	CLIENT_2	S3CO1, S3CO3, SRG1, SRG2
103	MGMT	S3CO1, S3CO3, SRG1, SRG2
999	L3_P2P	S3CO1, S3CO3

3. Configure all distribution switches (S3CO1-4) ports connect to S2CO1 and S2CO2 as trunk ports. Configure vlan 888 as native vlan across these trunk ports and allow all VLANs in the table. Don't forget to configure both S2COx devices.
 - (a) Also configure SRG1 and SRG2 G0/1 interface as trunk ports. You can leave vlan 1 as native vlan for these trunk ports.
4. Configure a logical interface (use 12 for interface number) between S3CO1 and S3CO2 to pass all vlan across it. Assign physical interface G1/0/23 and G1/0/24 to it. You require to use LACP as protocol and both switches should be able to initiate the negotiation.
5. Configure a logical interface (use 34 for interface number) between S3CO3 and S3CO4 without using any dynamic negotiation. Assign G1/0/23 and G1/0/24 physical interface into logical interface.
6. Configure HSRP for the 4 vlans (100-103) to meet following conditions.
 - (a) In each subnet, default gateway IP address needs to be .10 in each /24 subnet. Use vlan number as HSRP group number when configuring.
 - (b) S3CO1 should be the HSRP active for COR1 and S3CO3 should be the HSRP active for COR2 with priority 110 and leave the default value for non priority.
 - (c) In case of active device (S3CO1 or S3CO3) goes down S3CO2 or S3CO4 should act as active device in each site. When S3CO1 or S3CO3 comes back after failure, it should take over HSRP active role once it is operational.

EIGRP

Configure EIGRP protocol in COR2 (CO2, RCO3, RCO4, S3CO3 and S3CO4 devices) to achieve the following requirements.

1. For IPv4, use autonomous system number 100. Make sure loopback 1 IP address become EIGRP router-id.
2. Advertise all loopback and /30 P2P (point-to-point/) networks into EIGRP on CO2, RCO3, RCO4, S3CO3 and S3CO4. Advertise vlan 100-103 networks into EIGRP in S3CO3 and S3CO4. Please configure no auto-summary.

OSPF

Configure OSPF protocol in COR1 (CO1, RCO1, RCO2, S3CO1 and S3CO2 devices) to achieve the following requirements.

1. Configure OSPF process 100 in each of those devices. Use Loopback 1 interface as router-id in each of those devices.
2. Advertise all /30. P2P links and Loopback 1 interfaces to OSPF area 0.
3. S3CO1 and S3CO2, configure vlan 100-103 network in OSPF area 0.

STATIC ROUTING

1. Configure default routes on CO1, CO2, RG1 and RG2 routers pointing towards ISP1.
2. Configure static routes on:

(a) CO1 to reach COR2, REG1 and REG2 via CO2

```
ip route 10.11.0.0 255.255.0.0 192.0.2.202
ip route 10.1.0.0 255.255.0.0 192.0.2.202
ip route 10.2.0.0 255.255.0.0 192.0.2.202
```

(b) CO2 to reach REG1, REG2 and COR1 via CO1

```
ip route 10.10.0.0 255.255.0.0 192.0.2.102
ip route 10.1.0.0 255.255.0.0 192.0.2.2
ip route 10.2.0.0 255.255.0.0 192.0.2.6
```

3. Configure route redistribution:
 - (a) On CO1: redistribute static routes into OSPF process 100
 - (b) On CO2: redistribute static routes into EIGRP AS 100
4. Ensure that all end devices in VLAN networks across COR1, COR2, REG1 and REG2 can communicate with each other.

IP Services

You have been asked to configure following IP services.

1. When vlan 101-102 users (~400 in total users and require simultaneous internet access) in COR1, COR2, REG1 and REG2 communicate with internet (You can use 8.8.8.8 IP for testing) their addresses should be translated to following addresses depend on the internet router it goes through.

(a) Traffic goes via CO1 -> 192.0.2.104 -192.0.2.110

```
access-list 10 permit 10.10.101.0 0.0.0.255
access-list 10 permit 10.10.102.0 0.0.0.255
access-list 10 permit 10.11.101.0 0.0.0.255
access-list 10 permit 10.11.102.0 0.0.0.255
access-list 10 permit 10.1.101.0 0.0.0.255
access-list 10 permit 10.2.102.0 0.0.0.255

ip nat pool CO1_POOL 192.0.2.104 192.0.2.110 netmask 255.255.255.240

ip nat inside source list 10 pool CO1_POOL overload
```

(b) Traffic goes via CO2 -> 192.0.2.193 -192.0.2.199

```
access-list 20 permit 10.10.101.0 0.0.0.255
access-list 20 permit 10.10.102.0 0.0.0.255
access-list 20 permit 10.11.101.0 0.0.0.255
access-list 20 permit 10.11.102.0 0.0.0.255
access-list 20 permit 10.1.101.0 0.0.0.255
access-list 20 permit 10.2.102.0 0.0.0.255

ip nat pool CO2_POOL 192.0.2.193 192.0.2.199 netmask 255.255.255.240

ip nat inside source list 20 pool CO2_POOL overload
```

2. Configure DHCP services on S3CO1 and S3CO3 to meet requirements below.
on S3CO1
 - (a) VL101 DHCP address scope 10.10.101.101-10.10.101.254 with default router of 10.10.101.10
 - (b) VL102 DHCP address scope 10.10.102.101-10.10.102.254 with default router of 10.10.102.10**on S3CO3**
 - (c) VL101 DHCP address scope 10.11.101.101-10.11.101.254 with default router of 10.11.101.10
 - (d) VL102 DHCP address scope 10.11.102.101-10.11.102.254 with default router of 10.11.102.10
3. Configure DHCP services REG1 and REG2 in a way clients RG1-CL11 get IP address from vlan 101 and RG2-CL11 get IP address from vlan 102.
 - (a) REG1- VL101 DHCP address scope 10.1.101.101-10.1.101.254 with default-gateway 10.1.101.10.
 - (b) REG2- VL102 DHCP address scope 10.2.102.101-10.2.102.254 with default-gateway 10.2.102.10.

Security

1. Enable SSH on CO1 and CO2 routers to meet requirements given below.
 - (a) Use most secure SSH version.
 - (b) Use **admin/Skill39@ID** credential.
 - (c) Telnet should not be allowed when accessing network devices.
2. Limit only CO1-SRV (10.10.100.101/24) can SSH into CO1 and CO2 with the name ACL: **SSH_ONLY**.
3. Enable port security on CO1-CLI and CO2-CLI connected switchports.
 - (a) Limit two mac addresses.
 - (b) In case of port security violation, port should be disabled and syslog message to be generated.